EKNOOR KAUR SANDHU

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EDUCATION

SGGS Collegiate Public School, Chandigarh, CBSE — Expected May 2027

- Biology, Mathematics, Artificial Intelligence, Economics. Only student in the school with this combination.
- Grade 11: 95.8%, rank 2 of 371. Grade 12 midterms: 96.4%, rank 2 of 366. SAT 1510 (superscore).

RESEARCH

Silver Nanoparticle–Silk Fibroin Hydrogels, Research Intern, Plaksha University — Grade 12

- Made silver-loaded silk fibroin hydrogels from cocoons. Confirmed nanoparticles by UV-Vis. Tested against Gram-positive and Gram-negative bacteria.
- Result: 50 µg/mL matched a dose 500 times higher. Aim: lowest effective silver dose for wound dressings.
- First author. Journal of Emerging Investigators, submitted September 2026, under review.

Crop Imaging for Sugarcane, Plaksha University; DronaMaps (paid research intern) — Grades 11–12

- Grade 11: tested satellite imagery for sugarcane maturity. Dropped the claim: sucrose builds up inside the stalk and does not show in images. Presented at Punjab Agricultural University.
- Grade 12: flew survey drones and logged ground truth (archive about 1,790 acres) for a training-free, RGB-only row-gap tracker.
- 11 evaluated fields (about 5 ha): recovered 79% of 399 annotated rows. Standard method: 9%.
- Co-author, Dynamic Row Gap. Presented at ICA 2026 (International Conference on Agriculture-Centric Computation), Ropar.

Bioinformatics, Self-directed — Grades 11–12

- Online courses: Bioinformatics (UC San Diego), Genomic Data Science (Johns Hopkins).
- Compared fibroin sequences with BLAST, Clustal and T-Coffee at Sri Guru Gobind Singh College (access arranged with the principal's help). Taught sequence alignment to 30 classmates.

Independent Coding, Self-directed — Grades 9–12

- pdf-repeat-remover (Python; AP revision decks), dance-face-detect (TensorFlow; Kathak spins), po-di-ng (multiplayer Pong in Odin, TCP, no AI tools), fast-todo (FastAPI, SQLite).

LEADERSHIP

Head Girl, Student Council, SGGS Collegiate Public School — Grade 12 (Secretary, Grade 11)

- Nominated from all four senior streams. Won majorities in three.
- Documented timetable clashes and staffing gaps for management. School has begun appointing 2 AI teachers, 1 Economics teacher and 1 counsellor. Held orientation sessions for juniors.
- Handed sponsor contacts to the incoming Secretary. Team raised about USD 7,600 for MUN and the Annual Function.

President, Environment Club; Lead, Green Campus Programme, SGGS with APSWDP — Grades 11–12

- Set up student audits of classroom electricity, waste segregation and single-use plastic. Now run by students two years junior. APSWDP uses the model in 29 Chandigarh schools.
- Selected by APSWDP as youth representative, COP31, Antalya, November 2026.

President, Dance Club, SGGS Collegiate Public School

- Trained 17 dancers for school functions and 2 state-level competitions (wins: folk 2, Western 1). Organised 5 environmental street plays.

WORK AND SERVICE

Volunteer, Intern and Master Trainer, APSWDP, Chandigarh — Grade 9–present

- Supervised outreach support for women in sex work, transgender persons and other vulnerable communities. Worked with 9 peer educators and 31 field staff.
- Follow-up records: clinic referrals, HIV/STI testing, identity documents, child education, welfare schemes.
- Team results: 997 ABHA health IDs; identity documents for 150 transgender persons.

Entrepreneurship Intern, Punjab Innovation Mission — Grade 11

- Market research, competitor mapping and prototyping for 6 startup founders. Presented to 240 attendees.

Annotation Quality Intern, Labellerr AI — Grade 11

- Python audit scripts for COCO, Pascal VOC, JSON and CSV exports. Checked 477 files for missing labels and empty samples.

HONORS AND SCHOLARSHIPS

- **AP Scholar with Distinction**. 5s: Calculus BC, Microeconomics. 4s: Biology, Chemistry, CS A. Self-prepared — 2026
- **Cultural Talent Search Scholarship (Kathak)**. Ministry of Culture, India. 7 years of training — Grades 9–12
- **STEM Scholar**. Junior Academy, New York Academy of Sciences. Energy-retrofit team project — Grade 12
- **Young Technology Scholar**. Plaksha University. About 700 applicants, 437 schools — Grade 10

SKILLS AND INTERESTS

- **Technical:** Python, TensorFlow, Java, FastAPI, SQLite, Odin, BLAST, Clustal, T-Coffee, UV-Vis, drone surveys
- **Languages:** English, Punjabi, Hindi, French. **Interests:** Kathak, sketching, watercolour, stone painting